SHANI "DILL" DESILVA
U14033199

(1A)

ABCD Value	INPUT A	B	C	D	Output F
0	0	0	0	0	1
1	0	0	0	1	0
2	0	0	1	0	1
3	0	0	1	1	0
4	0	1	0	0	0
5	0	1	0	1	1
6	0	1	1	0	0
7	0	1	1	1	1
8	1	0	0	0	1
9	1	0	0	1	0
10	1	0	1	0	1
11	1	0	1	1	0
12	1	1	0	0	0
13	1	1	0	1	1
14	1	1	1	0	0
15	1	1	1	1	1

(1b) $\sum m$ of $F' = [1, 3, 4, 6, 9, 11, 12, 14]$

$(\prod M$ of $F = \sum m$ of $F')$

If "this function" means for F,

$\prod M$ of $F = [1, 3, 4, 6, 9, 11, 12, 14]$

If "this function" means F' in the previous sentence,

$\prod M$ of $F' = [0, 2, 5, 7, 8, 10, 13, 15]$.

added after asking Prof

The inverse of F is
$a'b'c'd + a'b'cd + a'bc'd' + a'bcd' + abc'd' + abcd' +$
$\hookrightarrow ab'c'd + ab'cd.$
which simplifies to $bd' + b'd$ using Kmap from 1c.

Shani 'DILL' De Silva
U1403199

Shani "Dill" De Silva
U1403199

(1c)

KMAP

AB \ CD	00	01	11	10
00	1 (0)	0 (1)	0 (3)	1 (2)
01	0 (4)	1 (5)	1 (7)	0 (6)
11	0 (12)	1 (13)	1 (15)	0 (14)
10	1 (8)	0 (9)	0 (11)	1 (10)

$B'D' + BD$

$a'b'c'd' + ab'c'd' + b'cd' + b'c'd + bcd + a'bd$

(0) (8) (2,10.) (5,13.) (7,15) (7,5.)

~~PIs~~	0	2	5	7	8	10
$a'b'c'd'$						

QUINE MCCLUCKEY Proof

EPIs.	0 $a'b'c'd'$	2 $a'b'cd'$	5 $a'bc'd$	7 $a'bcd$	8 $ab'c'd'$	10 $ab'cd'$	13 $abc'd$	15 $abcd$
BD			X	X			X	X
$B'D'$	X	X			X	X		

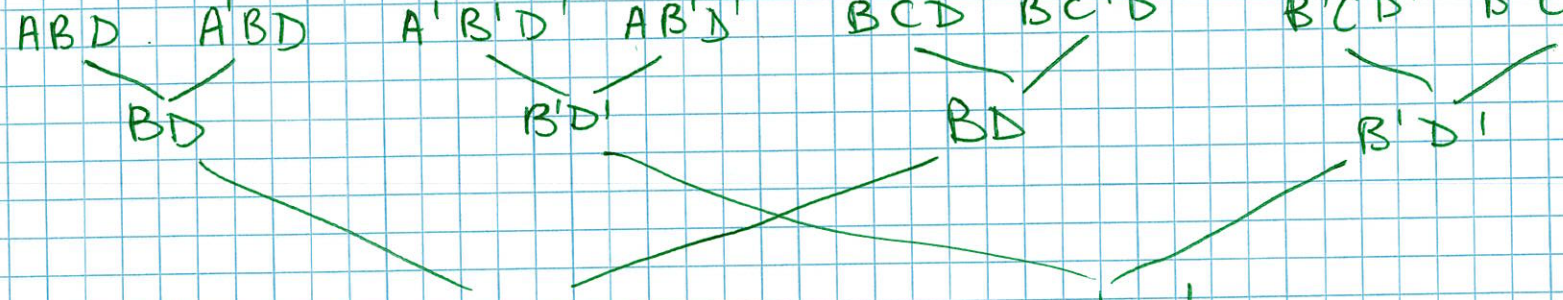
↑ from KMAP

Both BD & $B'D'$ are EPIs.

MINIMAL EXPRESSION = $BD + B'D'$.

1c, cont. QUINE MC CLUCKEY ABCD

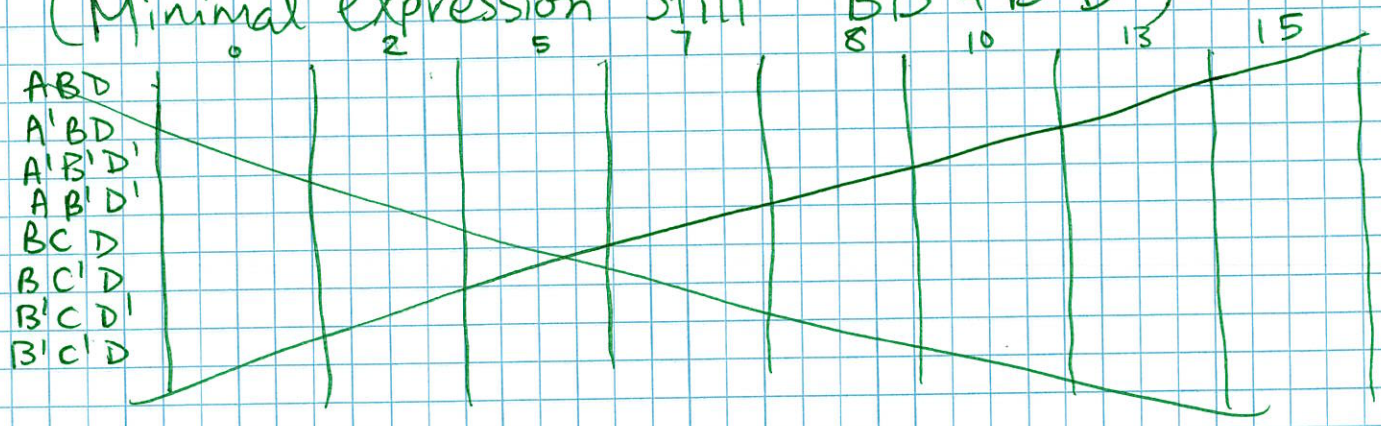
implicant options: ABD, A'BD, A'B'D', AB'D', ~~BCD~~, BC'D, B'CD', B'C'D'.



Quine McCluckey PIs: BD + B'D'.

PI	ab'c'd' (0)	a'b'cd' (2)	a'bc'd (5)	a'b'cd (7)	ab'cd' (8)	ab'cd' (10)	abc'd (13)	abcd (15)
BD	0	0	1	1	0	0	1	1
B'D'	1	1	0	0	1	1	0	0

(Minimal expression still BD + B'D')



Shani Dill De Silva
U1403319 9

Shani "Dill" De Silva
U14033199

(1d) EPIs: BD & $B'D'$.

	0	2	5	7	8	10	13	15
BD			X	X			X	X
$B'D'$	X	X			X	X		

Both BD and $B'D'$ are EPIs because they contain at least 1 unique X in a minterm column so they are required to cover that minterm.

BD is required to cover $[5, 7, 13, 15]$ and
$B'D'$ is required to cover $[0, 2, 8, 10]$.

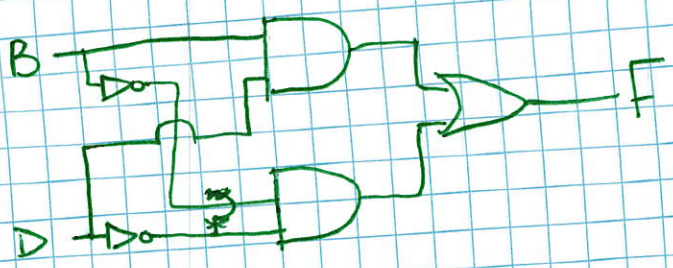
(1e) there are no static hazards in $BD + B'D'$ because there aren't any adjacent midterms not covered by the same EPI. no PIs need to be added.

Shani "Dill" De Silva
U14033199.

Q1F

Shani 'Dill' DeSilva
U1403199.

logic circuit.

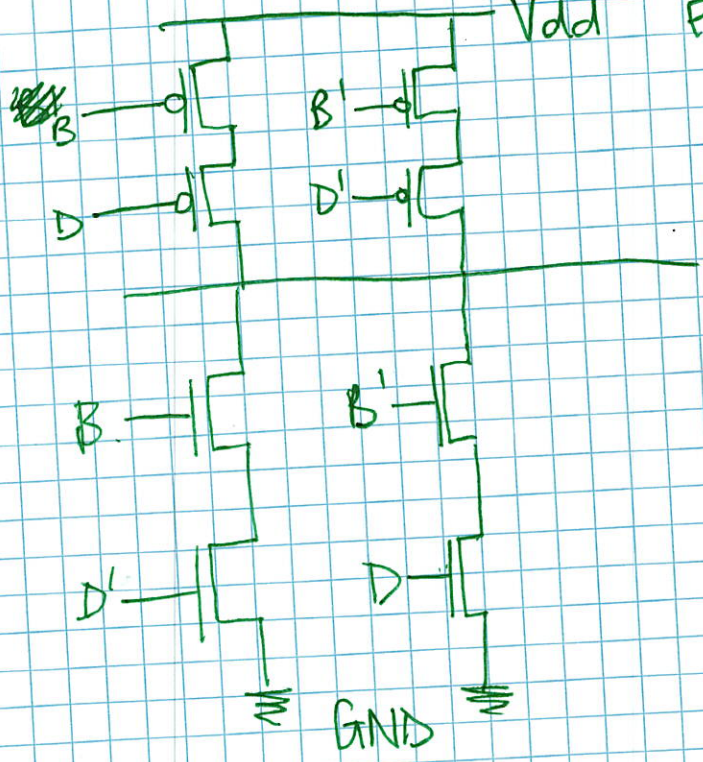


A, C } Not needed.

CMOS CIRCUIT:

$F = BD + B'D$

$F' = BD' + B'D.$



Shani "Dill" De Silva
U1403 3199

1G
1H
1I
1J

Answers typed & inserted into scan of answers.